Part A · Sparse Linear Predictors and Model Selection

Notation: $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{-1, +1\}$, $\mathcal{H}_k = \{x \mapsto \operatorname{sign}(\langle w, x\rangle) : \|w\|_0 \le k\}$, $\mathcal{H} = \cup_{k=1}^{d} \mathcal{H}_k$, and $S \sim \mathcal{D}^n$.

A.0 Requirement Mapping

- A1 VC bound via supports: Section A1.
- A2 penalty-SRM oracle inequality: Section A2.
- A3 sparse-vs-dense sample complexity comparison: Section A3.
- A4 validation model-selection bound and comparison: Section A4.
- A5 brute-force computation/runtime discussion: Section A5.

A1. VC dimension through supports

For each $I \subseteq [d]$ with $|I| = k$, define

$$\mathcal{H}_I = \{x \mapsto \operatorname{sign}(\langle w, x\rangle) : \operatorname{supp}(w) \subseteq I\}.$$

Then

$$\mathcal{H}_k = \bigcup_{I:|I|=k} \mathcal{H}_I, \qquad \#\{I : |I| = k\} = \binom{d}{k}.$$

Growth function:

$$\Pi_{\mathcal{H}_k}(n) \le \sum_{I:|I|=k} \Pi_{\mathcal{H}_I}(n) \le \binom{d}{k} \max_I \Pi_{\mathcal{H}_I}(n).$$

Each $\mathcal{H}_I$ is homogeneous halfspaces in $\mathbb{R}^k$, so by Sauer–Shelah

$$\Pi_{\mathcal{H}_I}(n) \le \sum_{j=0}^{k} \binom{n}{j} \le (en/k)^k.$$

Hence

$$\Pi_{\mathcal{H}_k}(n) \le \binom{d}{k} (en/k)^k \le (ed/k)^k (en/k)^k.$$

Setting $m = \operatorname{VCdim}(\mathcal{H}_k)$ and using $\Pi_{\mathcal{H}_k}(m) = 2^m$:

$$2^m \le (e^2 dm/k^2)^k$$

Taking logs:

$$m \log 2 \le k \log(e^2 d/k^2) + k \log m$$

so

$$\frac{m}{k} \le \frac{1}{\log 2} \big(\log(e^2 d/k^2) + \log m\big),$$

and the RHS grows only logarithmically in m, hence $m \le C\, k \log(ed/k)$ for a universal constant C. Therefore

$$\boxed{\mathrm{VCdim}(\mathcal{H}_k) = O\big(k \log(ed/k)\big).}$$

AI suggestion rejected/modified: - Rejected: cite a black-box union-VC lemma only. Modified to explicit growth-function steps to match the hint.

A2. Penalty-based SRM

Choose prior $(p_k)_{k=1}^d$, $p_k > 0$, $\sum_k p_k \le 1$. For each k, from A1:

$$d_k := \mathrm{VCdim}(\mathcal{H}_k) \le C_1 k \log(ed/k).$$

A class-level SRM bound gives (simultaneously for all $k, h \in \mathcal{H}_k$, w.p. $1-\delta$):

$$L_{\mathcal{D}}(h) \le L_S(h) + 2\sqrt{\frac{d_k + \log(1/p_k) + \log(1/\delta)}{n}},$$

where the constant 2 can be replaced by the lecture-version constant (absorbed later into C). Define

$$\mathrm{pen}(k, n, \delta) := 2\sqrt{\frac{C_1 k \log(ed/k) + \log(1/p_k) + \log(1/\delta)}{n}}.$$

Let

$$\hat{h} \in \arg\min_{1 \le k \le d,\ h \in \mathcal{H}_k} \{L_S(h) + \mathrm{pen}(k, n, \delta)\}.$$

Then on the same event:

$$\boxed{L_{\mathcal{D}}(\hat{h}) \le \inf_{1 \le k \le d,\ h \in \mathcal{H}_k} \left\{ L_{\mathcal{D}}(h) + C\sqrt{\frac{k \log(ed/k) + \log(1/p_k) + \log(1/\delta)}{n}} \right\}}$$

for a universal C.

Interpretation of costs: - $k \log(ed/k)$: statistical price for choosing support coordinates (which sparse directions). - $\log(1/p_k)$: extra price for choosing the sparsity level k across models.

AI suggestion rejected/modified: - Rejected: use $p_k \equiv 1/d$ without discussion. Modified to keep general p_k, as requested by the prompt.

A3. Comparison with dense halfspaces

Assume $\exists w^\star$ with $\|w^\star\|_0 \le s$ and $L_{\mathcal{D}}(w^\star) \le \eta$. Plug $k = s, h = w^\star$ in A2:

$$L_{\mathcal{D}}(\hat{h}) \le \eta + C\sqrt{\frac{s \log(ed/s) + \log(1/p_s) + \log(1/\delta)}{n}}.$$

To guarantee $L_{\mathcal{D}}(\hat{h}) \le \eta + \varepsilon$, it is sufficient that

$$n \;\ge\; \frac{C^2}{\varepsilon^2}\Big(s\log(ed/s) + \log(1/p_s) + \log(1/\delta)\Big).$$

For dense homogeneous halfspaces in $\mathbb{R}^d$, VC dimension is $\Theta(d)$, giving sample size

$$n = \Theta\left(\frac{d + \log(1/\delta)}{\varepsilon^2}\right).$$

Sparse SRM is smaller when

$$s\log(ed/s) + \log(1/p_s) \ll d.$$

In particular, if $p_k \propto 1/k^2$, then $\log(1/p_s) = O(\log s + \log \sum_{j=1}^{d} 1/j^2) = O(\log s)$, so the dominant term is $s\log(ed/s)$.

AI suggestion rejected/modified: - Rejected: hide $\log(1/p_s)$ inside $O(\cdot)$. Modified to keep constants/terms explicit.

A4. Validation as model selection

Split S into S_1, S_2 of size $n/2$. For each k, fit

$$w_k \in \arg\min_{\|w\|_0 \le k} L_{S_1}(w),$$

then select

$$\hat{k} \in \arg\min_{1 \le k \le d} L_{S_2}(w_k), \qquad \hat{w} := w_{\hat{k}}.$$

Condition on S_1: candidates $\{w_1, \dots, w_d\}$ are fixed, so validation is finite-class ERM over d hypotheses. With prob. $1 - \delta/2$ over S_2,

$$L_{\mathcal{D}}(\hat{w}) \le \min_k L_{\mathcal{D}}(w_k) + C_3\sqrt{\frac{\log(d/\delta)}{n/2}}.$$

Also, with prob. $1 - \delta/2$ over S_1, uniformly in k:

$$L_{\mathcal{D}}(w_k) \le \inf_{\|w\|_0 \le k} L_{\mathcal{D}}(w) + C_4\sqrt{\frac{k\log(ed/k) + \log(d/\delta)}{n/2}}.$$

Since $1/(n/2) = 2/n$, the extra factor $\sqrt{2}$ is absorbed into constants. Therefore, by a union bound, with prob. $1 - \delta$:

$$\boxed{L_{\mathcal{D}}(\hat{w}) \le \inf_{1 \le k \le d,\ \|w\|_0 \le k} \left\{ L_{\mathcal{D}}(w) + C\sqrt{\frac{k\log(ed/k) + \log(d/\delta)}{n}} \right\}.}$$

Comparison to penalty-SRM: - Penalty-SRM: same sample used for fitting and choosing k; pays $\log(1/p_k)$ model-index cost. - Validation: S_1 fits predictors, S_2 chooses k; pays finite-choice cost $\log d$ and a factor from sample split.

AI suggestion rejected/modified: - Rejected: treat $\{w_k\}$ as independent of S_1. Modified by conditioning on S_1 before applying finite-class bound.

A5. Computation in realizable case

Assume there exists a k-sparse consistent homogeneous halfspace for the sample.

Brute force algorithm: 1. Enumerate all supports $I \subseteq [d]$, $|I| = k$ (there are $\binom{d}{k}$). 2. Restrict each sample x_i to coordinates in I, obtaining points in $\mathbb{R}^k$. 3. Solve feasibility: find $u \in \mathbb{R}^k$ such that $y_i \langle u, x_i^{(I)} \rangle > 0$ for all i (or non-strict with margin slack/LP reformulation). 4. If feasible, lift u back to $w \in \mathbb{R}^d$ with support I, output w.

If feasibility per support costs $T(n,k)$ (polynomial in n,k), total runtime is

$$O\left(\binom{d}{k} T(n,k)\right) \approx O\big((ed/k)^k \, T(n,k)\big) \, .$$

So it is polynomial in d when $k = O(1)$, quasi-polynomial/subexponential for $k = \Theta(\log d)$, and exponential in d when $k = \Theta(d^\alpha)$ ($\alpha > 0$).

This illustrates Week 4: sample complexity can be small (depends on $k \log(ed/k)$) while exact computation can still be intractable due to support search.

AI suggestion rejected/modified: - Rejected: call runtime "polynomial" for all k. Modified to explicit dependence via $\binom{d}{k}$.

A6. Traceability pass for Part A

- A1 support-union and growth-function combination: done in A1.
- A2 SRM oracle inequality with instantiated terms and interpretation: done in A2.
- A3 sparse-vs-dense sample-size comparison and regime statement: done in A3.
- A4 validation oracle inequality and comparison with penalty-SRM: done in A4.
- A5 brute-force algorithm, runtime, and computational-vs-statistical discussion: done in A5.

References: Sauer (1972), Shelah (1972), Hoeffding (1963), McDiarmid (1989).

Part B · PAC-Bayes for Thresholds

We use the notation from the prompt:

$$\mathcal{X}_N = \{1, 2, \ldots, N\}, \quad \mathcal{Y} = \{0, 1\}, \quad \mathcal{H}_N = \{h_t : t \in \{1, \ldots, N{+}1\}\}, \quad h_t(x) = \mathbf{1}[x \geq t].$$

In Part B we are in the realizable setting.

B.0 Requirement Mapping (assignment bullet → section)

- B1 (VC dimension and point posterior): Sections B1.1-B1.4.
- B2 (version space form, size, KL, empirical risk, interpretation): Sections B2.1-B2.5.
- B3 (risk formula for uniform posterior on W, concrete example, why no contradiction): Sections B3.1-B3.4.
- B4 (fixed prior attack, probability event, unique consistent threshold, KL lower bound, interpretation): Sections B4.1-B4.6.

B1. VC dimension and point-posterior PAC-Bayes

B1.1 Prove $\mathrm{VCdim}(\mathcal{H}_N) = 1$

Lower bound ≥ 1: a single point $\{x\}$ can be labeled both ways by choosing $t = x$ (label 1) and $t = x + 1$ (label 0).

Upper bound ≤ 1: no two-point set can be shattered. Let $x_1 < x_2$. For any threshold classifier, if $h_t(x_1) = 1$, then automatically $h_t(x_2) = 1$. So labeling $(1, 0)$ on (x_1, x_2) is impossible. Therefore no size-2 set is shattered.

Hence

$$\boxed{\mathrm{VCdim}(\mathcal{H}_N) = 1.}$$

B1.2 Point posterior KL

Let P be uniform on $\mathcal{H}_N$, so

$$P(h_t) = \frac{1}{N+1} \quad \forall t.$$

Let ERM return a consistent $\hat{h}_t$, and set

$$Q_S = \delta_{\hat{h}_t}.$$

Then

$$\mathrm{KL}(Q_S \| P) = \sum_h Q_S(h) \log \frac{Q_S(h)}{P(h)} = \log \frac{1}{P(\hat{h}_t)} = \log(N+1).$$

B1.3 Plug into PAC-Bayes bound

Using the lecture bound in the prompt,

$$L_{\mathcal{D}}(Q) \leq L_S(Q) + \sqrt{\frac{\mathrm{KL}(Q\|P) + \log \frac{2n}{\delta}}{2(n-1)}}.$$

For $Q_S = \delta_{\hat{h}_t}$, realizability plus consistency gives $L_S(Q_S) = 0$, hence

$$\boxed{L_{\mathcal{D}}(Q_S) \leq \sqrt{\frac{\log(N+1) + \log \frac{2n}{\delta}}{2(n-1)}}.}$$

B1.4 Why this scales with $\log(N+1)$

The point posterior commits all mass to one threshold, so the KL term pays a coding cost of identifying one hypothesis among $N+1$ possibilities, namely $\log(N+1)$. VC-style realizable guarantees for thresholds depend on VC dimension $= 1$, not on the domain size N, so they do not carry this $\log N$ dependence.

B1.5 AI suggestion rejected/modified

- Rejected suggestion: state $\mathrm{VCdim}(\mathcal{H}_N) = 1$ without proving non-shatterability of two points.
- Reason: the assignment asks for a proof, so the monotonic-threshold argument on $x_1 < x_2$ is written explicitly.

B2. Version-space posterior

Define

$$a(S) = \max\{x_i : y_i = 0\},\ a(S) = 0 \text{ if no negatives},$$

$$b(S) = \min\{x_i : y_i = 1\},\ b(S) = N+1 \text{ if no positives}.$$

B2.1 Characterize consistent thresholds

A threshold h_t is consistent iff it predicts 0 on every negative and 1 on every positive.

For a negative point x_i (label 0), consistency means

$$h_t(x_i) = \mathbf{1}[x_i \geq t] = 0 \iff x_i < t.$$

So we must have $t > a(S)$.

For a positive point x_i (label 1), consistency means

$$h_t(x_i) = \mathbf{1}[x_i \geq t] = 1 \iff x_i \geq t.$$

So we must have $t \leq b(S)$.

Combining both,

$$V(S) = \{t : a(S) < t \leq b(S)\}.$$

Therefore

$$|V(S)| = b(S) - a(S).$$

Edge cases are consistent with this formula: if there are no negatives then $a(S) = 0$ and only $t \leq b(S)$ is required; if there are no positives then $b(S) = N+1$ and only $t > a(S)$ is required.

B2.2 Uniform posterior on version space

Let

$$Q_V(h_t) = \begin{cases} \frac{1}{|V(S)|}, & t \in V(S), \\ 0, & t \notin V(S). \end{cases}$$

Prior is uniform: $P(h_t) = 1/(N+1)$.

Then

$$\mathrm{KL}(Q_V \| P) = \sum_{t \in V(S)} \frac{1}{|V(S)|} \log \frac{1/|V(S)|}{1/(N+1)} = \log \frac{N+1}{|V(S)|}.$$

B2.3 Show $L_S(Q_V) = 0$

Every h_t with $t \in V(S)$ is consistent with S, so $L_S(h_t) = 0$. Hence

$$L_S(Q_V) = \mathbb{E}_{h \sim Q_V}[L_S(h)] = 0.$$

B2.4 Why this can improve the PAC-Bayes certificate

For point posterior $\delta_{\hat{h}}$: $\mathrm{KL} = \log(N+1)$. For version posterior Q_V: $\mathrm{KL} = \log((N+1)/|V(S)|)$. If $|V(S)|$ is large, then

$$\log \frac{N+1}{|V(S)|} \ll \log(N+1),$$

so the complexity term in PAC-Bayes is smaller.

B2.5 AI suggestion rejected/modified

- Rejected suggestion: claim $|V(S)| = b(S) - a(S) + 1$.
- Reason: thresholds are indexed by integers with strict/weak inequalities $a(S) < t \leq b(S)$, so the exact count is $b(S) - a(S)$.

B3. Spreading helps KL, but can hurt true risk

Assume realizable $\mathcal{D}$: $X \sim \mathrm{Unif}(\{1, \ldots, N\})$, and labels are generated by h_τ. For nonempty $W \subseteq \{1, \ldots, N+1\}$, let Q_W be uniform on $\{h_t : t \in W\}$.

B3.1 Compute $L_{\mathcal{D}}(Q_W)$

For fixed t, $L_{\mathcal{D}}(h_t)$ equals disagreement probability between h_t and h_τ:

$$L_{\mathcal{D}}(h_t) = \frac{1}{N} \left| \{x \in \{1, \ldots, N\} : h_t(x) \neq h_\tau(x)\} \right|.$$

If $t \geq \tau$, disagreement occurs on $x \in \{\tau, \ldots, t-1\}$, count $t - \tau$. If $t < \tau$, disagreement occurs on $x \in \{t, \ldots, \tau - 1\}$, count $\tau - t$. So in all cases,

$$L_{\mathcal{D}}(h_t) = \frac{|t - \tau|}{N}.$$

Averaging under Q_W:

$$L_{\mathcal{D}}(Q_W) = \frac{1}{|W|} \sum_{t \in W} L_{\mathcal{D}}(h_t) = \frac{1}{N|W|} \sum_{t \in W} |t - \tau|.$$

This is exactly the required formula.

B3.2 Concrete example ($N \geq 20$)

Take

$$N = 20, \quad \tau = 11.$$

Take realizable sample

$$S = \{(1, 0), (20, 1)\}.$$

This is realizable by h_{11} since $1 < 11 \Rightarrow y = 0$, $20 \geq 11 \Rightarrow y = 1$.

For this sample,

$$a(S) = 1, \quad b(S) = 20,$$

so

$$V(S) = \{2, 3, \ldots, 20\}, \quad |V(S)| = 19$$

(large).

KL comparison under uniform prior on $N + 1 = 21$ thresholds:

$$\mathrm{KL}(Q_{V(S)} \| P) = \log \frac{21}{19},$$

while

$$\mathrm{KL}(\delta_{h_\tau} \| P) = \log 21.$$

Hence

$$\mathrm{KL}(Q_{V(S)} \| P) < \mathrm{KL}(\delta_{h_\tau} \| P).$$

Risk comparison:

$$L_{\mathcal{D}}(\delta_{h_\tau}) = 0$$

(since h_τ is Bayes/realizable truth), but

$$L_{\mathcal{D}}(Q_{V(S)}) = \frac{1}{20 \cdot 19} \sum_{t=2}^{20} |t - 11| = \frac{90}{380} = \frac{9}{38} > 0.$$

So the required strict inequality holds.

B3.3 Why this does not contradict PAC-Bayes

PAC-Bayes gives an upper bound on true risk, not a guarantee that lower KL implies lower true risk for every posterior. A spread posterior can reduce KL but still average in many inaccurate hypotheses, increasing $L_{\mathcal{D}}$. No contradiction occurs because the theorem compares each posterior to its own bound; it does not totally order posteriors by true risk via KL alone.

B3.4 AI suggestion rejected/modified

- Rejected suggestion: use a non-realizable sample to make $|V(S)|$ large.
- Reason: the question explicitly requires a realizable sample, so the example was chosen to be consistent with h_τ.

B4. Every fixed prior can be attacked

Let any prior P over $\mathcal{H}_N$ be fixed before data; assume $N \geq 3$.

B4.1 Find an internal threshold with small prior mass

There are $N-1$ internal thresholds $\{2, \ldots, N\}$. Their masses sum to at most 1:

$$\sum_{t=2}^{N} P(h_t) \leq 1.$$

By averaging/pigeonhole, there exists

$$\tau \in \{2, \ldots, N\} \quad \text{such that} \quad P(h_\tau) \leq \frac{1}{N-1}.$$

B4.2 Define adversarial realizable distribution

Define $\mathcal{D}_\tau$ on two points:

$$\Pr[(x,y) = (\tau-1, 0)] = \frac{1}{2}, \qquad \Pr[(x,y) = (\tau, 1)] = \frac{1}{2}.$$

This is realizable by threshold h_τ.

B4.3 Probability sample contains both support points

For $S \sim \mathcal{D}_\tau^n$, event "miss $(\tau-1, 0)$" has probability $(1/2)^n$. Same for "miss $(\tau, 1)$". By union bound,

$$\Pr[\text{at least one support point missing}] \leq 2 \cdot \left(\frac{1}{2}\right)^n = 2^{1-n}.$$

Therefore

$$\Pr[\text{both support points appear}] \geq 1 - 2^{1-n}.$$

B4.4 On this event, $V(S) = \{\tau\}$

If both points appear, consistency requires: - $h_t(\tau-1) = 0 \Rightarrow \tau - 1 < t$, - $h_t(\tau) = 1 \Rightarrow t \leq \tau$. Hence

$$\tau - 1 < t \leq \tau.$$

Since t is integer, uniquely $t = \tau$. So

$$V(S) = \{\tau\}.$$

B4.5 Consequence for zero-empirical-error posteriors and KL

If $L_S(Q) = 0$, every hypothesis in support of Q must be consistent with all sample points. But on this event only h_τ is consistent, so

$$Q = \delta_{h_\tau}.$$

Justification of the first sentence: $L_S(Q) = \mathbb{E}_{h\sim Q}[L_S(h)] = 0$ with $L_S(h) \geq 0$ for all h, so any h with $Q(h) > 0$ must satisfy $L_S(h) = 0$ (otherwise the expectation would be strictly positive). Then

$$\mathrm{KL}(Q\|P) = \log \frac{1}{P(h_\tau)} \geq \log(N-1).$$

So on that event, for every zero-training-error posterior,

$$\boxed{\mathrm{KL}(Q\|P) \geq \log(N-1).}$$

B4.6 What this does and does not show

It shows a limitation of fixed-prior, zero-empirical-error PAC-Bayes certificates: for some realizable distributions, the KL term is inevitably large. It does not show a sample-complexity lower bound for learning thresholds; thresholds still have VC dimension 1 and are learnable with sample complexity independent of N (up to ε, δ terms). The result is about certificate tightness under this proof template, not impossibility of learning.

B4.7 AI suggestion rejected/modified

- Rejected suggestion: conclude this proves thresholds need $\Omega(\log N)$ more samples.
- Reason: that conclusion is false; the argument lower-bounds a PAC-Bayes KL certificate for fixed prior and zero training error, not the true PAC sample complexity.

B5. Traceability pass for Part B

- B1 VC dimension proof and point-posterior PAC-Bayes: done in B1.1-B1.4.
- B2 version-space form, cardinality, exact KL, and $L_S(Q_V) = 0$: done in B2.1-B2.3.
- B2 interpretation (why version posterior can improve bound): done in B2.4.
- B3 exact formula for $L_{\mathcal{D}}(Q_W)$: done in B3.1.
- B3 concrete $N \geq 20$ example satisfying both inequalities: done in B3.2.
- B3 explanation of no contradiction: done in B3.3.
- B4 existence of internal τ with small prior mass: done in B4.1.
- B4 high-probability event $\geq 1 - 2^{1-n}$: done in B4.3.

- B4 uniqueness $V(S) = \{\tau\}$, zero-error posterior characterization, and KL lower bound: done in B4.4-B4.5.
- B4 interpretation (what it does/does not show): done in B4.6.

References: Hoeffding (1963), McDiarmid (1989), PAC-Bayes lecture theorem.

Part C · AI Usage Report

I used AI in both Part A and Part B. I used it to check algebra and bound forms, and to help type and format my write-up. I treated AI as a collaborator, not as an oracle.

AI suggestions were used across the whole process: drafting structure, checking intermediate derivations, and revising for completeness. I ran follow-up checks on contradiction steps, sample-splitting factors, and edge cases.

One AI suggestion I rejected was in Part A.3: an earlier version hid terms such as $\log(1/p_s)$ inside big-O notation too early. I rejected that and kept these terms explicit to match the assignment requirement.

I manually verified all final content before submission. In particular, I checked inequality transitions, sample-splitting denominators, edge cases, and sign/operator correctness in equations.

Process improvements from this assignment

1. Add a proof-completeness checklist prompt after each AI draft: list every inequality/bound step and cite the theorem used.
2. Add a constant/normalization tracking step: explicitly track scaling factors and constants through to the final bound.
3. Add an edge-case checklist for set/interval characterizations (for example, no-positive or no-negative samples).
4. Separate derivation and interpretation passes: one pass for math steps, one pass for "what this shows vs. what it does not show."
5. Add a final sign/operator check to ensure all $+, \leq, \geq$ symbols are present and correct.